Ans-2-a

LSTM stands for a Long Short-Term Memory, and it's a special type of Recurrent Neural Network (RNN) designed to learn long term dependencies in sequential data like time series, speed or text. It was introduced by Hochreiter & Schmidhuber in 1997 to solve the limitations of traditional RNNs.

LSTM networks overcome the key limitation of tradition traditional Recurrent Neural Network (RNN), particularly the problem of vanishing gradients. In standard RNNs, as the network backpropogates through many timesteps, the gradients can become extremely small, making it difficult for model to learn long-term dependencies. LSTM address this by introducing a special memory cell that can maintain information over long durations. This cell is regulated by three gates - the forget gate, input gate and ~~op~~ output gate which control flow of information into, out of, and within the cell. These gates allow the network to selectively remember or forget information ensuring that important features are retained across long sequences. As a result, LSTMs are more effective than traditional RNNs of learning from long sequences and modelling complex temporal differences

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best